

POLITECNICO DI TORINO



Lab 4

Exercise 1 - Ping

Music and multimedia streaming over the Internet

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Full Name : MEHRAN BAYAT

Student Number : S320636.

Cristina Rottondi, Leonardo Severi, Massimiliano Zanoni

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README

Lab 4 Exercise 1: Ping

ICMP Echo Request and Echo Reply Analysis Using Wireshark

Crownlabs VM, Terminal, and Wireshark Capture

Final_Report.md

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1. Title

ICMP Ping Analysis Using Terminal and Wireshark in the Crownlabs Lab 2 Virtual Machine

2. Abstract

In this report, I explain what I did for **Lab 4 - Exercise 1: Ping**. In this exercise, I had to choose one known IP address, send exactly three ping packets to it, and then check the generated packets in Wireshark.

For my test, I selected the IP address **130.192.9.250**. I started the capture in Wireshark on the **enp1s0** interface because the traffic was going outside the VM. Then I used this command in the terminal:

```
ping -c 3 130.192.9.250
```

In Wireshark, I could see three ICMP Echo Request packets going from my VM to the selected IP address and three ICMP Echo Reply packets coming back. My VM address in the capture was **172.16.203.155**. The result confirmed that the selected host replied correctly to all three ping requests.

The main point I understood from this experiment is that **ping** uses **ICMP over IPv4**. It does not use TCP and it does not use UDP.

3. Exercise Objective

The objective of this exercise was to see what happens at packet level when I use the **ping** command. The exercise was not only about running the command. I also had to inspect the packet capture and identify the protocols and packet directions.

The task I performed can be summarized like this:

```
Choose one IP address -> run ping with exactly 3 packets -> capture the traffic in  
Wireshark -> identify Echo Request and Echo Reply packets
```

For the deliverable, I had to report:

1. the IP address I chose;
2. the exact command I used;
3. the protocols involved;
4. one Echo Request or Echo Reply packet;
5. the packet type, source IP address, and destination IP address.

4. Lab Environment and Tools

4.1 Working Environment

I performed the experiment inside the **Crownlabs Lab 2 virtual machine**. I used the terminal to run the ping command and Wireshark to capture and inspect the packets.

Tool	How I used it
Terminal	I used it to execute the <code>ping</code> command.
Wireshark	I used it to capture the packets and check their source, destination, protocol, and type.

4.2 Wireshark Capture Interface

For this exercise, I used the `enp1s0` interface. I used this interface because I was pinging an external IP address, not `localhost`. The loopback interface would be correct for local traffic, but here the packet had to leave the VM through the virtual Ethernet interface.

4.3 Capture Filter

Before starting the capture, I used the filter required by the lab instructions:

```
not tcp port 5901 and not tcp port 6080 and not tcp port 80 and not tcp port 443
```

This filter was useful because it reduced unrelated traffic from the remote VM connection and web traffic. It did not block ICMP traffic, so the ping packets were still captured correctly.

5. Ping Command Execution

5.1 Chosen IP Address

From the list of IP addresses in the lab, I chose:

```
130.192.9.250
```

5.2 Exact Command Used

The command I executed was:

```
ping -c 3 130.192.9.250
```

The option `-c 3` is important because it makes the command send exactly three ping requests. Without this option, the ping command would continue running until I stopped it manually.

5.3 Meaning of the Command

Part of command	Meaning
<code>ping</code>	Starts the reachability test.
<code>-c 3</code>	Sends exactly three Echo Request packets.
<code>130.192.9.250</code>	The destination IP address that I selected.

6. Protocols Involved

6.1 Main Protocols

The main protocols that appeared in the useful packets were:

Protocol	Role in my capture
Ethernet II	This was the lower layer shown by Wireshark because I captured packets from a network interface.
IPv4	This carried the ICMP packets between my VM and the destination IP address.
ICMP	This was the main protocol used by <code>ping</code> ; it generated the Echo Request and Echo Reply messages.

For the direct lab answer, the most important protocols are **IPv4** and **ICMP**. Wireshark also shows Ethernet II because it displays the lower layers of the packet as well.

6.2 Why TCP and UDP Were Not Used

This exercise did not use TCP or UDP. I understood this because `ping` does not open a TCP connection and does not send UDP datagrams. Instead, it sends ICMP messages directly inside IPv4 packets.

Because of this, I did not see TCP flags such as `SYN`, `ACK`, or `FIN`, and I also did not see UDP source and destination ports for the ping packets.

7. Wireshark Capture Result

7.1 Capture Screenshot

After running the command, I filtered the useful packets in Wireshark and focused on the packets related to `130.192.9.250`. The capture showed the three request/reply pairs clearly.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	fe80::a8bb:aaff:fe49:abb5	ff02::2	ICMPv6	70	Router Solicitation from aa:bb:aa:49:ab:b5
2	44.035466349	fe80::c858:86ff:febb:f40a	ff02::2	ICMPv6	70	Router Solicitation from ca:58:86:bb:f4:0a
3	110.851635296	172.16.203.155	130.192.9.250	ICMP	98	Echo (ping) request id=0x0001, seq=1/256, ttl=64 (reply in 4)
4	110.852530635	130.192.9.250	172.16.203.155	ICMP	98	Echo (ping) reply id=0x0001, seq=1/256, ttl=61 (request in 3)
5	111.853026704	172.16.203.155	130.192.9.250	ICMP	98	Echo (ping) request id=0x0001, seq=2/512, ttl=64 (reply in 6)
6	111.853425198	130.192.9.250	172.16.203.155	ICMP	98	Echo (ping) reply id=0x0001, seq=2/512, ttl=61 (request in 5)
7	112.878542006	172.16.203.155	130.192.9.250	ICMP	98	Echo (ping) request id=0x0001, seq=3/768, ttl=64 (reply in 8)
8	112.878856776	130.192.9.250	172.16.203.155	ICMP	98	Echo (ping) reply id=0x0001, seq=3/768, ttl=61 (request in 7)

Frame 3: 98 bytes on wire (784 bits), 98 bytes captured on interface enp1s0
enp1s0: <stopped capture> Packets: 8 - Displayed: 8 (100.0%)

Figure 1: Wireshark capture of the ping experiment. The useful packets are the ICMP Echo Requests from 172.16.203.155 to 130.192.9.250 and the Echo Replies in the opposite direction.

7.2 Observed Packet Sequence

The important part of the capture was the ICMP packet sequence. I sent three requests and received three replies:

No.	Source	Destination	Protocol	Information
3	172.16.203.155	130.192.9.250	ICMP	Echo Request, sequence 1
4	130.192.9.250	172.16.203.155	ICMP	Echo Reply, response to request 1
5	172.16.203.155	130.192.9.250	ICMP	Echo Request, sequence 2
6	130.192.9.250	172.16.203.155	ICMP	Echo Reply, response to request 2
7	172.16.203.155	130.192.9.250	ICMP	Echo Request, sequence 3
8	130.192.9.250	172.16.203.155	ICMP	Echo Reply, response to request 3

This is what I expected: every Echo Request had one Echo Reply. So the result shows that the selected host was reachable from my VM.

7.3 Background Packets Observed

In the capture I also saw some ICMPv6 Router Solicitation packets. These packets were not part of my ping command to 130.192.9.250. They were just background traffic from the VM network environment.

For this reason, I ignored the ICMPv6 packets and focused on the IPv4 ICMP packets involving the selected IP address. A useful display filter for this was:

```
icmp && ip.addr == 130.192.9.250
```

8. Selected Packet for the Deliverable

8.1 Selected Packet

For the deliverable, I selected the first ICMP Echo Request packet from the capture. In my packet list, this was packet number 3.

Required field	Captured value
Packet type	Echo Request
Source IP address	172.16.203.155
Destination IP address	130.192.9.250
Protocol	ICMP over IPv4

8.2 My Interpretation

I chose an Echo Request packet because it clearly shows the direction from my VM to the selected destination. The source IP address was 172.16.203.155, which was the address of my VM in the capture. The destination was 130.192.9.250, which was the IP address I selected from the lab list.

The next packet was the Echo Reply, going from 130.192.9.250 back to 172.16.203.155. This confirms that the remote host answered my ping request.

9. Direct Answers for the Lab Deliverable

This section contains the exact answers that I would put in the Lab 4 deliverable template for 4.1.1 To known IP address.

9.1 Ping to Known IP Address

1. **Chosen IP:** 130.192.9.250
2. **Exact command issued:** `ping -c 3 130.192.9.250`
3. **List of protocols involved:** Ethernet II, IPv4, ICMP. For the main lab answer, the important ones are IPv4 and ICMP.

9.2 Chosen Echo Packet

1. **Packet type:** Echo Request
2. **Source IP address:** 172.16.203.155
3. **Destination IP address:** 130.192.9.250

10. Explanation of the Result

The ping test was successful. My VM sent three ICMP Echo Request packets to 130.192.9.250, and the destination answered with three ICMP Echo Reply packets.

This also explains why Wireshark showed six useful ICMP packets even though the command used `-c 3`. The command sends three request packets, but Wireshark captures both directions:

```
1 Echo Request + 1 Echo Reply = 1 complete ping exchange
3 ping exchanges = 3 Echo Requests + 3 Echo Replies = 6 ICMP packets
```

So the packet count in Wireshark is correct. It includes the packets sent by my VM and the packets received back from the destination.

11. Short Comparison with the Named-Host Ping Part

In this completed part, I pinged a known IP address directly. Because the destination was already an IP address, my VM did not need to ask DNS to translate a name.

If I had used a hostname such as `leonardo.polito.it`, then DNS packets would appear before the ICMP packets. That is the main difference between pinging a known IP address and pinging a named host. In the known-IP part documented here, DNS was not needed.

12. Final Summary

In this exercise, I selected `130.192.9.250`, ran `ping -c 3 130.192.9.250`, captured the traffic on `enp1s0`, and identified the useful ICMP packets in Wireshark.

The final result was:

Item	Final result
Chosen IP	130.192.9.250
Command	<code>ping -c 3 130.192.9.250</code>
Interface	<code>enp1s0</code>
Main protocols	IPv4 and ICMP
Requests captured	3 ICMP Echo Requests
Replies captured	3 ICMP Echo Replies
Selected packet	Echo Request
Source IP of selected packet	172.16.203.155
Destination IP of selected packet	130.192.9.250

The exercise confirmed that `ping` uses ICMP Echo Request and Echo Reply packets. It also helped me see clearly how Wireshark shows packet direction, source IP, destination IP, and protocol.

References

1. Lab 4 assignment PDF, *Music and multimedia streaming over the Internet*, Politecnico di Torino, Academic year 2025/2026.
2. Lab 4 deliverable template, Section 4.1.1, *To known IP address*.

3. Wireshark packet capture performed in the Crownlabs Lab 2 virtual machine.